

KARLEBY-JAKOBSTAD, FinlandWMO: **029030**Lat: **63.718N**Lon: **23.139E**Elev: **83**StdP: **14.65**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.1	-7.9	-20.1	1.8	-14.9	-12.6	2.8	-8.3	23.5	28.4	21.2	29.1	1.8	30	0.667

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	80.3	64.8	76.8	63.0	73.2	61.2	67.3	76.0	65.2	73.1	63.3	70.7	9.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.4	90.9	70.3	62.4	84.4	68.9	60.0	77.5	66.4	31.7	76.4	30.1	72.8	28.7	70.7	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.4	18.0	16.0	DB	-20.2	84.3	8.3	4.6	-26.2	87.6	-31.0	90.3	-35.7	92.9	-41.7	96.2	(4)
(5)			WB	-21.6	69.2	8.8	3.1	-27.9	71.4	-33.0	73.2	-38.0	75.0	-44.3	77.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.9	19.1	18.9	25.6	36.6	46.4	55.1	61.5	57.9	49.5	38.7	30.8	25.0	(6)	
	DBStd	16.96	12.71	12.82	9.59	6.70	7.74	6.43	6.22	5.93	6.41	7.23	8.65	11.04	(7)	
	HDD50	4972	958	871	756	403	162	22	7	86	353	577	775		(8)	
	HDD65	9590	1423	1291	1221	851	576	305	145	231	465	815	1027	1240	(9)	
	CDD50	911	0	0	0	3	51	174	357	252	71	3	0	0	(10)	
	CDD65	54	0	0	0	0	1	7	35	11	0	0	0	0	(11)	
	CDH74	616	0	0	0	0	35	91	375	115	0	0	0	0	(12)	
	CDH80	113	0	0	0	0	2	20	75	16	0	0	0	0	(13)	
(14) Wind	WSAvg	7.3	7.3	7.4	7.6	7.4	7.8	7.6	6.5	6.5	6.9	7.2	7.7	7.6	(14)	
(15) Precipitation	PrecAvg	21.6	1.3	1.1	1.1	1.0	1.7	1.9	2.8	2.7	2.0	2.3	1.9	1.8	(15)	
	PrecMax	28.6	2.8	2.1	2.2	2.6	3.1	3.8	4.8	5.7	4.3	4.6	4.1	4.2	(16)	
	PrecMin	15.4	0.4	0.1	0.2	0.2	0.3	0.6	0.6	1.0	0.6	0.6	0.3	0.4	(17)	
	PrecStd	3.2	0.6	0.6	0.5	0.6	0.8	0.9	1.1	1.3	1.1	0.9	0.9	0.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.6	42.6	48.6	64.7	78.9	82.5	86.2	82.2	70.5	57.1	50.1	43.9	(19)	
		MCWB	38.6	38.3	40.3	50.1	59.4	64.9	68.1	64.7	60.7	52.8	48.0	42.8	(20)	
	2%	DB	37.6	38.8	44.2	57.9	73.0	75.5	82.0	77.1	66.2	53.7	46.8	41.3	(21)	
		MCWB	36.0	36.1	38.5	46.7	56.7	60.4	66.4	64.0	58.2	51.0	45.3	40.0	(22)	
	5%	DB	35.7	35.8	40.6	53.3	66.6	71.8	78.4	73.4	62.8	51.6	44.5	39.4	(23)	
		MCWB	34.6	34.2	36.1	44.2	53.5	58.4	64.4	62.2	56.2	49.3	43.3	38.1	(24)	
	10%	DB	33.9	33.9	37.5	48.7	61.2	68.4	74.9	69.8	60.4	49.6	41.7	37.2	(25)	
		MCWB	33.3	32.8	34.3	41.6	50.9	57.1	63.2	61.1	55.3	47.9	40.7	36.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.7	38.6	41.7	52.4	61.4	66.7	71.3	69.6	62.0	54.1	48.3	42.9	(27)	
		MCDB	39.4	41.4	47.0	60.6	75.0	77.1	81.3	76.9	67.9	56.3	49.5	43.2	(28)	
	2%	WB	36.2	36.1	38.7	48.2	58.2	63.4	68.3	66.1	59.5	51.7	45.4	40.2	(29)	
		MCDB	37.1	38.0	42.4	56.6	69.7	72.5	77.5	72.7	64.7	53.6	46.6	40.8	(30)	
	5%	WB	34.5	34.5	36.6	44.8	54.9	60.5	66.4	63.9	57.5	49.8	43.0	38.1	(31)	
		MCDB	35.2	35.5	39.5	51.4	65.6	68.8	75.1	70.3	61.5	51.1	43.8	39.0	(32)	
	10%	WB	33.1	32.9	34.7	42.0	51.7	58.4	64.5	62.0	56.0	47.7	40.7	35.9	(33)	
		MCDB	33.7	33.7	36.9	48.0	60.0	66.3	72.4	68.3	59.4	48.9	41.4	36.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	12.5	14.7	16.7	19.8	18.0	18.5	17.9	15.8	10.8	8.1	9.6	(35)	
		MCDBR	8.4	9.8	15.8	24.8	29.9	25.1	25.2	23.8	19.6	11.2	8.7	8.4	(36)	
	5% WB	MCWBR	8.0	8.3	11.2	15.3	15.9	12.7	11.9	12.4	12.5	8.8	8.1	7.6	(37)	
		MCWBR	8.7	8.8	13.3	22.1	26.8	21.7	21.8	19.8	17.1	10.2	8.4	8.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.227	0.287	0.314	0.352	0.350	0.351	0.371	0.358	0.333	0.313	0.364	0.352	(40)	
		taud	2.021	2.157	2.225	2.294	2.440	2.472	2.438	2.470	2.521	2.377	2.626	2.582	(41)	
	Ebn at Noon		115	198	245	261	274	276	266	259	243	193	106	58	(42)	
		Edh at Noon	9	20	28	33	31	30	31	27	21	16	8	5	(43)	
(44) All-Sky Solar Radiation	RadAvg		44	224	701	1204	1625	1794	1640	1252	718	268	62	19	(44)	
	RadStd		4	44	74	108	166	154	154	121	74	49	10	2	(45)	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page